

Sree Lekha Kolla

☎ 217-761-6923 | ✉ sreelekhakolla957@gmail.com | 💼 Sree Lekha Kolla | 🌐 GitHub | 📍 San Jose, California

Education

Master of Science, Data Analytics | GPA:3.90

Graduated: May 2025

University of Illinois Springfield | Springfield, Illinois

Bachelor of Technology, Information Technology | GPA:3.30

Graduated: May 2023

Gayatri Vidya Parishad College of Engineering for Women | Visakhapatnam, India

Skills and Certifications

Certifications AWS Cloud Practitioner,
Data Analytics with Python [IIT Roorkee]

Others Tableau, Power BI, MS Office, Django, React,
AWS, OOPS, GIT, JIRA, Excel.

Databases MySQL, Oracle, PostgreSQL

QA & Testing Manual Testing, Functional & Regression

Languages Python, C++, Java, R, SQL, Javascript

Testing, Web Application Testing, UI/UX Test

Cloud AWS (EC2, S3, SNS, Glue, Lambda, VPC, Auto
Scaling, Load Balancing)

Case Writing, Bug Reporting & Tracking

Experience

Software Engineer

Feb 2024 - May 2025

Innovate Springfield

- Designed and implemented a scalable Member Management System with **React** and **Django/MySQL**, ensuring modular architecture and high availability.
- Built and deployed **RESTful APIs** for frontend-backend integration and inter-service communication, following microservices design principles.
- Optimized **database queries and indexing in MySQL**, improving response times by 35% for high-traffic operations.
- Performed **ETL** and built real-time dashboards with **SQL**, **Tableau**, and **Power BI**, supporting operational analytics.
- Developed **forecasting models (Random Forest, ARIMA)** to predict member engagement and revenue, improving planning and resource allocation.

Cloud Support Associate

Jul 2022 - Jul 2023

PricewaterhouseCoopers Acceleration Centers

- Designed and deployed scalable cloud infrastructure on **AWS (VPC, EC2 with auto-scaling and load balancing)**.
- Automated backend operations with **AWS Lambda** and **SNS**, improving efficiency and uptime.
- Optimized web apps by integrating **AWS S3** for static content delivery (20% faster load times).
- Performed **manual/automated functional testing** for cloud-hosted applications and backend services.
- Implemented **ETL pipelines in Python** for data processing and cross-system integration.

Programmer Intern

Jan 2022 - May 2022

Enlignce Technologies

- Developed a full-stack interview preparation platform using **Python**, **Django**, and **MySQL**.
- Implemented unit tests to validate **CRUD operations** across roles and ensure reliability.
- **Optimized relational database schemas** and implemented **indexing strategies** to enhance query performance.
- Built **RESTful APIs** for question banks, candidate submissions, and evaluation services, enabling modular design.
- Integrated **speech-to-text APIs** for automated spoken answer evaluation (reduced manual work by 40%).

Projects

Natural Language Processing(NLP) tool for Neurodivergent users

- Built a **full stack NLP** app with **speech-to-text**, **text-to-speech**, and **sentiment analysis**, enhancing accessibility for neurodivergent users aligning with **WCAG accessibility standards**, and applied NLP for **emotion classification**.
- Developed a **Chrome extension** for adaptive **text summarization (BART)** for dynamic summary length.
- Integrated **assistive technology APIs (Web Speech API, SpeechRecognition, gTTS)** to enhance usability for neurodivergent individuals.

Deep Learning: Weather Image Classification

- Built and trained deep learning models (**CNN, ResNet, Vision Transformers**) for 2D weather image classification, achieving over 96% accuracy using **transfer learning** and **data augmentation**.

Deep Learning: Leaf Disease Detection system

- Designed a **CNN** model with **ROI extraction** to detect leaf diseases, achieving 94% accuracy.
- Enhanced performance with preprocessing and **data augmentation**, and built a **Django-based website** for real-time detection. Published in the **UGC-approved journal IJIRT - Leaf Disease Detection**.